MIL-E-1/1374D
2 February 1976
SUPERSEDING
MIL-E-1/1374C
5 October 1971

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 3B24WB

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Rectifier, half-wave, high-vacuum

See figure 1

Mounting position: Any

Weight: 2.25 ounces (63.8 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epx	ib	Io	Barometric pressure, reduced
Unit:	V	kv	ma	mAdc	mmHg
Maximum:	5.25	20	300	60	225
Minimum:	4.75	---	---	---	---
Maximum:	2.63	20	150	30	225
Minimum:	2.38 (Note 1)	---	---	--- (Note 1)	---
TEST CONDITIONS:	5.0 Vac	---	---	---	---

GENERAL:

Qualification - Required

(D) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 1								
1301	Filament current	1, 2	Ef = 2.5 Vac	0.65	II	If	2.75	3.25	Aac
---	Gas	3	Pp = 50 W (min); t = 30	0.65	II	ΔIb	---	5	mAdc
1353	Operation of rectifiers	3	t = 60; epx = 20 kv	0.65	II	Io	60	---	mAdc
1231	Peak emission (1)	-	eb = 1,000 v (max)	0.65	II	is	950	---	ma
1231	Peak emission (2)	2	Ef = 2.5 Vac; eb = 1,000 v (max)	0.65	II	is	475	---	ma
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	-	Rp = 2,000 ohms; Ebb Ib = 15 mAdc	---	---	Ep	---	300	mVac
1041	Shock	-	Hammer angle = 30°; no voltages	---	---	---	---	---	---
---	Mechanical resonance	4		---	---	---	---	---	---
1256	Electrode current (anode)	-	Eb = 250 Vdc	---	---	Ib	135	---	mAdc
	Quality conformance inspection, part 3								
---	Life test	-	Group C: epx = 20 kv; Io = 60 mAdc (min); ib = 300 ma (min); t = 1,000 hours	---	---	---	---	---	---
---	Life-test end point:								
1231	Peak emission (1)	-		---	---	is	750	---	ma
1506	Heater-cycling life	5	Ef = 6.0 V	---	---	---	---	---	---
1002	Barometric pressure, reduced	6	225 mmHg; epx = 20 kv	---	---	---	---	---	---
1031	Vibration fatigue	6	Ef = 2.5 Vac; Eb Ib = 30 mAdc	---	---	---	---	---	---

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NOTES:

1. Using one-half of filament.
2. Test each half of filament separately.
3. The following tests shall be the first tests performed after the holding period in the following order: (1) Operation of rectifiers, (2) Gas.
4. The fundamental frequency of the cold filament shall not be between 10 and 165 Hz. The fundamental frequency of all other cold parts shall not be below 500 Hz. In performing the test the tube shall be vibrated with sinusoidal motion.
5. Heater-cycling life test shall be in accordance with test method 1506, except cycle = 1 minute on, 4 minutes off.
6. (D) This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 80
DSA - ES

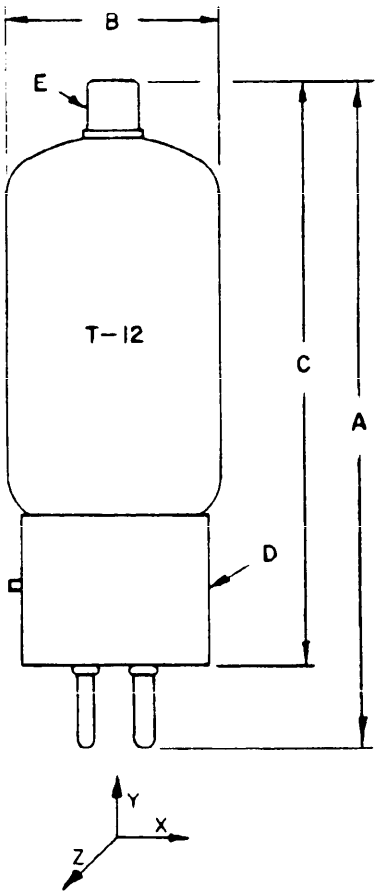
User activities:
Army - MU, WC
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3031)

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PIN CONNECTIONS	
Pin No.	Element
1	fct
2	f
3	nc
4	f
Cap	a

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A		4.812(122.22)
B		1.562 (39.67)
C	3.875(98.43)	4.125(104.78)
Ⓓ Quality conformance inspection, part 3 (see note b)		
D	Base: A4-10 (see note a) (EIA)	
E	Cap: C1-1 (EIA)	

NOTES:
a. Base material shall be low-loss phenolic.
b. Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 3B24WB.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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